FX-4CR

Multi-mode HF Software-Defined Transceiver
Technical Specifications and Operating Manual



Firmware version: R_2.0 (2026-10-01)
Applicable models: FX-4CR / FX-4R (identical firmware, different case marking)
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This manual combines the technical specifications and the operating instructions in one volume.
It is written for the person who operates the radio. All descriptions follow the actual panel silkscreen and the contents shown on the display; the quoted figures are those of the settings held in the present firmware.

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Part One - Technical and Functional Specifications

1 General description

The FX-4CR is a multi-mode software-defined (SDR) transceiver covering the HF amateur bands and the 6 m band, with wideband receiving capability on the medium-wave and short-wave broadcast bands as well. The radio uses digital quadrature sampling and fully digital signal processing: the received signal is digitally down-converted, digitally filtered and digitally demodulated before the audio is output. The transmit path supports phone, CW and data modes, and includes automatic level control, closed-loop output-power control and SWR detection.



Figure 1 Front of the transceiver (320 x 240 colour display, 11 function keys, 2 push-encoders, internal loudspeaker)

Main functions and configuration:

- Operating modes: CW (CW-L / CW-U), LSB, USB, LSB-D (data), USB-D (data), AM and FM - eight in all;
- Frequency range: 3.5-54 MHz in transceiver state (bands divided according to the selected ITU region), 522 kHz-30 MHz in receiver state;
- Dual VFO (VFO-A / VFO-B), with same-frequency and split operation, A/B switch and VFO / channel-mode switch;
- 100 channels (0-99); the VFO and channel memories of the transceiver state and of the receiver state are stored separately - eight memory groups in all;
- Nine tuning steps (1 Hz to 1 MHz); the frequency knob offers fast step selection while held and turned;
- Receiver: three-state front end (pre-amplifier / bypass / attenuator), AGC (slow / medium / fast) with manual gain (MGC), three noise-reduction levels (NR), pulse-noise blanker (NB), squelch (SQL), CW automatic decode, real-time spectrum and waterfall display;
- Transmitter: SSB / CW / AM / FM / data transmission, output power set and remembered per band, live display of output power and SWR;
- CW: straight key, Iambic A and Iambic B keying, dot/dash swap, sidetone pitch / volume / keying-speed adjustment, keyer practice mode and CW decode display;
- Control interfaces: USB-C serial link and Bluetooth wireless link, both compatible with the Kenwood TS-590S computer-control command set;

- All frequencies, modes, channels and settings are saved automatically and are retained when the power is removed.

2 Frequency range and band plan

The radio has two operating states: *transceiver* and *receiver* (broadcast). They are switched by holding the AM.FM key. In transceiver state the band edges depend on the ITU region selected in menu item 10 (1 / 2 / 3); the factory default is region 3.

Table 1 Tuning range of each amateur band in transceiver state, by ITU region

Band	Region 1 (China) MHz	Region 2 (US) MHz	Region 3 (Europe, default) MHz	Default MHz
80 m	3.500-3.800	3.500-4.000	3.500-3.900	3.800
60 m	5.330-5.3665	5.330-5.405	5.330-5.3665	5.360
40 m	7.000-7.200	7.000-7.300	7.000-7.200	7.050
30 m	10.100-10.150	10.100-10.150	10.100-10.150	10.100
20 m	14.000-14.350	14.000-14.350	14.000-14.350	14.270
17 m	18.068-18.168	18.068-18.168	18.068-18.168	18.100
15 m	21.000-21.450	21.000-21.450	21.000-21.450	21.200
12 m	24.890-24.990	24.890-24.990	24.890-24.990	24.900
10 m	28.000-29.700	28.000-29.700	27.000-29.700	29.200
6 m	50.000-54.000	50.000-54.000	50.000-54.000	50.200

Note: tuning past the edge of the current band enters the neighbouring band automatically. Transmit permission for the 60 m and 6 m bands can be enabled or disabled separately by the start-up key combinations (see Part Two, section 2). The small figures printed at the lower right of the panel keys (3.5, 5, 7, 10, 14, 18, 21, 24, 28, 50) are the start frequencies (MHz) of these bands and identify the band shortcuts.

Table 2 Transmit-permitted sub-bands

Band	Transmit range	Band	Transmit range
80 m	3.500-3.900 MHz	17 m	18.068-18.168 MHz
60 m	5.330-5.3665 MHz	15 m	21.000-21.450 MHz
40 m	7.000-7.200 MHz	12 m	24.890-24.990 MHz
30 m	10.100-10.150 MHz	10 m	28.000-29.700 MHz
20 m	14.000-14.350 MHz	6 m	50.000-54.000 MHz

Inside a band but outside the sub-band above, the radio receives only and will not transmit; in split operation the transmit frequency is subject to the same limit.

2.1 Receiver state: medium-wave and short-wave broadcast bands

The receiver state covers medium wave (MW) and 14 short-wave broadcast metre bands - 15 bands in all - from 522 kHz to 30 MHz.

Table 3 Broadcast bands in receiver state

Band	Range	Default	Band	Range	Default
MW	522-1710 kHz	945 kHz	25 m	11.600-13.100 MHz	12.000 MHz

120 m	2.300-2.498 MHz	2.400 MHz	22 m	13.570-13.870 MHz	13.700 MHz
90 m	3.200-3.400 MHz	3.300 MHz	19 m	15.100-15.800 MHz	15.500 MHz
75 m	3.900-4.000 MHz	3.950 MHz	16 m	17.480-17.900 MHz	17.700 MHz
60 m	4.700-5.600 MHz	5.500 MHz	13 m	21.450-21.750 MHz	21.500 MHz
49 m	5.900-7.000 MHz	6.000 MHz	11 m	25.600-27.405 MHz	27.000 MHz
41 m	7.100-7.800 MHz	7.200 MHz	10 m	28.000-30.000 MHz	29.000 MHz
31 m	9.400-9.900 MHz	9.500 MHz			

In receiver state the medium-wave default tuning step is 9 kHz; transmission is not possible in this state.

3 Operating modes

Table 4 List of operating modes

Display	Mode	Description
CW L	CW-LSB	CW, carrier beat inserted on the lower (low-frequency) sideband
CW U	CW-USB	CW, carrier beat inserted on the upper (high-frequency) sideband
LSB	Lower sideband	Phone mode, normally used below 10 MHz
USB	Upper sideband	Phone mode, normally used above 10 MHz
LSB_D	Lower sideband, data	Data mode, fixed 3.0 kHz bandwidth
USB_D	Upper sideband, data	Data mode, fixed 3.0 kHz bandwidth
AM	Amplitude modulation	Double-sideband AM phone / broadcast
FM	Frequency modulation	Narrow-band FM phone

When the radio is switched from CW to SSB for the first time, the sideband is chosen from the frequency: above 9.5 MHz the radio selects USB, below 9.5 MHz it selects LSB. After that, LSB / USB / LSB-D / USB-D can be cycled through freely.

4 Tuning steps

Table 5 Tuning-step selections

No.	Step	No.	Step
1	1 Hz	6	100 kHz
2	10 Hz	7	1 MHz
3	100 Hz	8	5 kHz
4	1 kHz	9	9 kHz (medium-wave default)
5	10 kHz		

A short press of the frequency knob swaps quickly between 1 kHz and 100 Hz. Holding the frequency knob and turning it cycles through all nine steps; releasing the knob returns to tuning. Changing the step rounds the frequency to a multiple of the new step automatically. In the main frequency display the digit that belongs to the current step is shown in yellow, and digits below the resolution of the step are shown dim.

5 VFO, channels and memory

- **Dual VFO:** VFO-A and VFO-B each store their own complete state - frequency, mode, filter, tuning step and so on - and can be switched with one key;
- **Split operation:** when enabled, one VFO frequency is used for receiving and the other for transmitting; during transmission the main and sub frequency displays swap places automatically;
- **VFO / channel mode:** in channel mode the frequency knob selects the channel number (0-99), and the complete current state can be written into a chosen channel;
- **Eight independent memory areas:** transceiver-state VFO-A / VFO-B / CH-A / CH-B and receiver-state VFO-A / VFO-B / CH-A / CH-B, all separate from one another;
- **Each channel stores:** frequency, band, mode, receive and transmit filter, tuning step, RIT/XIT, split setting, power, squelch, CW sidetone pitch / volume / keying speed, CW keyer type, NR / NB state, front-end attenuator / amplifier, IF gain, MIC gain and speech compression;
- All menu settings, panel states and calibration data are written to non-volatile memory automatically and are restored at the next start-up.

6 Digital signal processing and audio parameters

Table 6 DSP and audio parameters

Item	Parameter / specification
RF sampling rate	96 kHz (quadrature I/Q sampling)
DSP processing rate	24 kHz (digital IF; adjustable from 20.0 to 28.0 kHz in the menu)
CW decode processing rate	12 kHz
Audio A/D and D/A conversion	16-bit quantisation, 16-times oversampling on the receive channel
Frequency reference	25 MHz temperature-compensated crystal oscillator, calibration range 24.900000-27.100000 MHz
Demodulation	Fully digital for CW / SSB (including data modes) / AM / FM
Transmit modulation	Fully digital generation of SSB, CW, AM and FM
CW sidetone and beat frequency	Produced by a digital oscillator; sidetone pitch equals the receive beat frequency (500-990 Hz)
Spectrum analysis	1024-point FFT (512 bins), 320 display columns

7 Receiver functions

7.1 Front-end amplification and attenuation

A short press of the IF.ATT key cycles the front end through its states. Current hardware has three states, earlier hardware has two:

Table 7 Front-end amplifier / attenuator states

State	Current hardware (three states)	Earlier hardware (two states)
0	Pre-amplifier switched in (AMP ON)	Pre-amplifier switched in (AMP ON)
1	Amplifier and attenuator both bypassed (through)	Attenuator switched in (ATT ON)
2	Attenuator switched in (ATT1 / ATT2)	-

The hardware version is recognised automatically at start-up and the number of states follows it. The status bar shows ATT / ATT1 / ATT2 for the current front-end state; the indication is dim when no attenuation is applied.

7.2 AGC and manual gain (MGC)

- **AGC:** three recovery times - slow (S), medium (M), fast (F) - selected by menu item 5; the default is medium (M). The current setting appears as a single letter at the right of the status bar;
- **MGC manual gain:** holding the IF.ATT key switches between AGC (automatic) and MGC (manual). In MGC the IF gain is set in the MGC window of the right-hand parameter column, range 1-32;
- The manual gain value is stored per band.

7.3 Noise reduction (NR) and pulse blanker (NB)

The receiver has three digital noise-reduction levels and one pulse-noise blanker. The two are independent and can be on at the same time. Noise reduction works adaptively: once a level is selected the radio follows the signal and noise conditions by itself, and no strength or time-constant value has to be set.

Table 8 Digital noise-reduction (NR) levels

Level	Display	Function and where it is used
0	NR (dim)	Reduction off; the raw demodulated audio is passed to the output
1	NR1 (red)	Mild reduction: suppresses steady background noise while preserving the natural sound and the detail of speech; suitable for weak-signal phone and everyday contacts
2	NR2 (red)	Deep reduction: stronger suppression of broadband hiss (fans, generators, switch-mode supplies, line noise); the output sounds quieter than NR1 but less natural; suitable when the noise is heavy and speech is hard to pick out
3	NOT (red)	Automatic notch: removes a steady single-frequency interferer (carrier, whistle, howl, single-signal neighbour) while leaving the rest of the audio untouched; CW and data signals, which are themselves single tones, are protected automatically and are not notched away

Table 9 Pulse-noise blanker (NB)

Item	Description
Function	Suppresses the clicking produced by broadband pulse interference from ignition systems, sparking, appliance switches and the like
Switching	In phone modes, hold the M.V key to turn it on or off; the NB indication in the status bar turns red when the blanker is on
Where it applies	Only on the phone (LSB / USB) receive path; in CW mode the same key switches the CW decoder on and off
Relation to NR	Completely independent; both may be on together. Usually NB is enabled first to remove the pulses, and NR is then used on the remaining steady noise

Noise reduction is used in LSB / USB and in the data modes; AM and FM are not reduced. The status bar and the noise-reduction window use the same wording, so the indication cycles through NR / NR1 / NR2 / NOT.

7.4 Squelch (SQL)

The squelch threshold ranges from 0 to 99, default 0 (off). Audio is opened while the signal is above the threshold and closed, with a release delay, while it is below it, which removes the background noise between transmissions; the function is most often used in FM and when monitoring broadcast stations. The squelch value appears in the SQL box at the left of the screen and is adjusted by holding the AF knob.

7.5 CW automatic decode

Table 10 CW decode parameters

Item	Parameter
Processing rate	12 kHz
Detection method	Multi-tone sliding detection, 5 tones spaced 75 Hz apart, covering about +/-150 Hz of beat tuning
Speed range	Adaptive keying speed; correct decode from about 10 to 50 WPM and above
Decode buffer	128 characters, scrolling display
Shown	Decoded characters, measured keying speed (WPM) and signal level
Switching	Menu item 14 (CW_DEC), or hold the M.V key in CW mode
Display position	The area below the spectrum at the bottom of the screen (shared with the waterfall)

8 Digital filter selections

Table 11 Receive digital filter selections

Mode	Selections (bandwidth)	Default
CW	50 / 100 / 200 / 300 / 500 / 800 Hz (6)	500 Hz
SSB (LSB / USB)	1.5 / 1.8 / 2.1 / 2.4 / 2.7 / 3.0 kHz (6)	2.1 kHz
Data (LSB-D / USB-D)	Fixed 3.0 kHz	3.0 kHz
AM	6.0 / 9.0 kHz (2)	6.0 kHz
FM	5.0 / 10.0 kHz (2, corresponding to about 2.5 / 5 kHz deviation)	5.0 kHz

The SSB transmit filter is chosen separately in menu item 7 (TX_FILTER), six selections covering 1.5-3.0 kHz, defaulting to the 2.7 kHz selection. The receive bandwidth in use is shown in the box directly above the spectrum.

9 Transmitter functions

Table 12 Transmitter functions

Item	Parameter
Transmit modes	LSB / USB / LSB-D / USB-D / CW-L / CW-U / AM / FM
Power setting range, HF bands	0.1-20.0 W (0.1 W display resolution)
Power limit, 6 m band	5.0 W
Power limit, data modes (LSB-D / USB-D)	10.0 W
Power adjustment increment	0.1 W while the setting is 1.0 W or less; 1.0 W above 1.0 W
Power memory	The set power is stored separately for every band
Power control	Closed-loop output-power control (ALC) holds the set value
CW keying shaping	Output power ramps up and down, removing key clicks
Power / SWR display	Output power (W) and SWR are shown live during transmission; the SWR value is colour-coded by range
Receive level display	Shown both as a dBm figure and as the signal bar (S and dBm correspond; one bar segment is about 1 dB)

Supply voltage display	The supply voltage (V) is shown continuously in the status bar

Table 13 Phone transmission parameters

Item	Range	Default	Description
MIC gain	1-99	20	Microphone sensitivity; set in a window reached with the AF knob
Speech compression (TX_AF_COMP)	1-99	10	Menu item 8; strength of the transmit speech processing
Transmit SSB bandwidth	6 selections	2.7 kHz	Menu item 7, TX_FILTER
ALC start threshold	0-15	15	Developer menu item 6 (calibration)
ALC maximum gain	0-63	32	Developer menu item 7 (calibration)

10 CW (Morse) functions

Table 14 CW functions

Item	Parameter / range	Default
Keying method	Straight key (CW M), Iambic A (CW A), Iambic B (CW B)	Straight key
Keyer connection	CW.KEY socket, two inputs: a single straight key or a paddle with separate dot and dash sides	-
Dot / dash swap	Hold the CW key in CW mode to exchange the paddle directions	Normal
Practice mode	Held on or off with the RIT key in CW mode; sidetone only, no change to transmit	Off
Keying speed	5-50 WPM	20 WPM
Sidetone pitch (beat)	500-990 Hz in 10 Hz steps	650 Hz
Sidetone volume	0-32	1
CW hold-over delay	0-99 in 10 ms steps (0-990 ms)	5 (50 ms)
CW receive filter	50 / 100 / 200 / 300 / 500 / 800 Hz	500 Hz
CW sideband	CW-L / CW-U, switched by holding the SSB key in CW mode	CW-L
CW decode	Adaptive from about 10 to over 50 WPM, 128-character buffer	On

Note: the CW hold-over delay is the time the transmitter stays keyed after the key is released, which avoids switching back and forth between characters and words. In straight-key mode either keyer line or the PTT produces transmission; in automatic-keyer mode the dots, dashes and their spacing are generated from the paddle state.

11 Spectrum and waterfall display

Table 15 Spectrum / waterfall display parameters

Item	Parameter
Display area	Lower half of the screen: real-time spectrum (top) and waterfall (below)
Span selections	2: about 48 kHz / about 24 kHz (menu item 11, SPE_W)
Frequency resolution	About 150 Hz per column at 48 kHz; about 75 Hz per column at 24 kHz
Span corner marks	The waterfall corners show the half span: 24k or 12k
Grid	The centre vertical is the receive centre frequency; the side verticals are +/-20 kHz (48 kHz span)

FFT points	1024 points (512 bins), 320 display columns
Waterfall rows	50 rows, scrolling from bottom to top
Colour schemes	7 (menu item 12, SPE_COLOR, 0-6): 0 / 1 are colour gradients (filled / outline), 2-6 are single-colour outline (blue / yellow / green / red / grey)
Display amplitude	1-10 (menu item 13, SPE_AMP, default 4)
Transmit spectrum	Developer menu item 4 enables spectrum monitoring while transmitting

12 External connectors

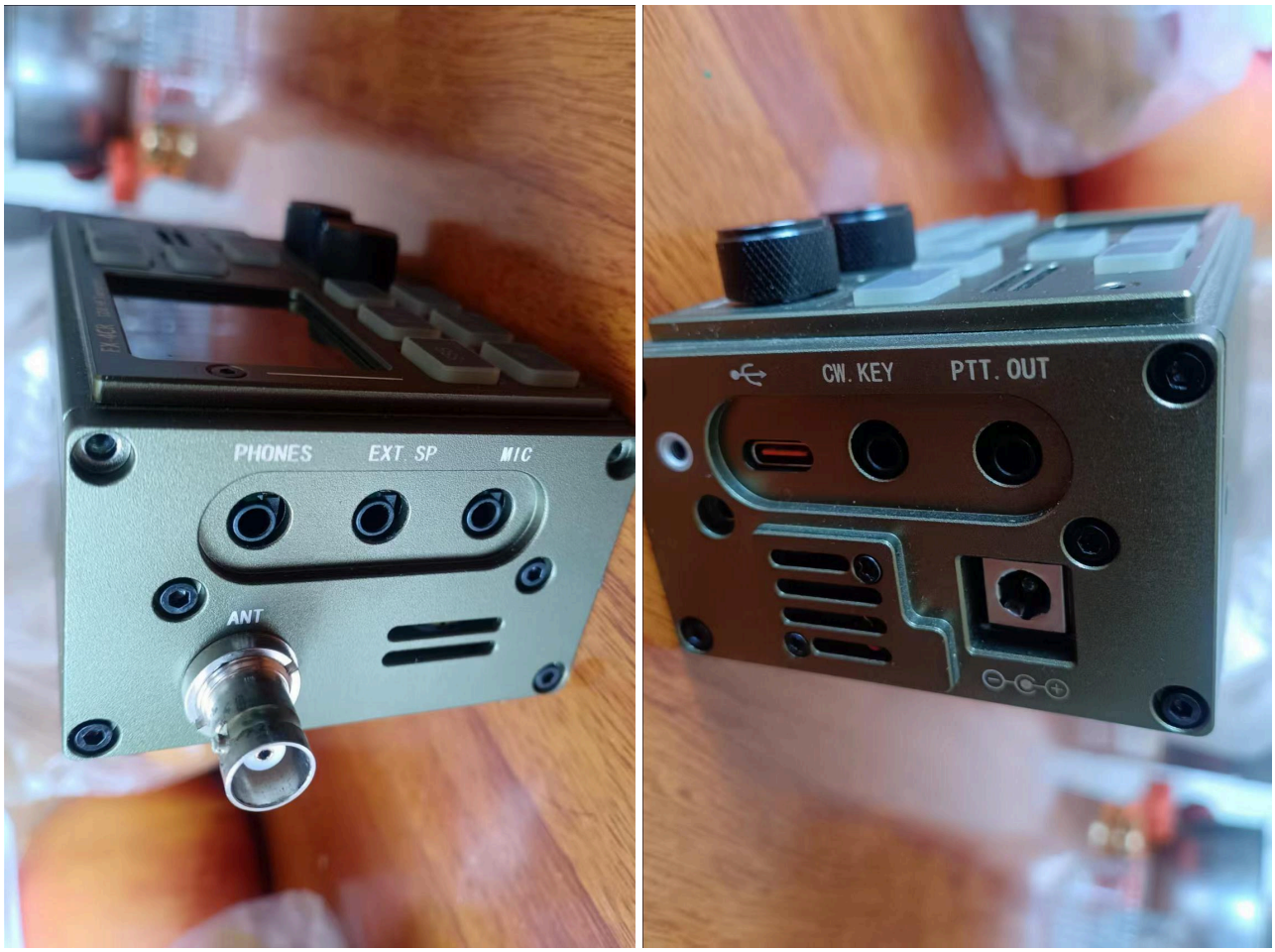


Figure 2 The two connector faces (left: ANT antenna socket, PHONES, EXT.SP loudspeaker output, MIC; right: PTT.OUT, CW.KEY, USB-C, d.c. power socket and ground screw)

Table 16 External connectors

Connector / marking	Description
ANT	RF antenna socket (BNC), used for both receiving and transmitting
PHONES	Headphone output; the internal loudspeaker is muted automatically when a plug is inserted
EXT.SP	External loudspeaker output
MIC	Microphone input for a hand microphone (including its PTT line); microphone bias is supplied
PTT.OUT	Transmit-control **output**: it operates while the radio is transmitting, to switch an external amplifier, antenna changer or similar device. This socket is **not** an external PTT input
CW.KEY	CW keyer input, two lines: a single straight key, or a paddle with separate dot and dash sides

USB-C	Data interface: CAT control and digital modes with a computer. The radio is **not** powered from USB
D.C. power socket	DC 12 V, centre positive
Ground screw	Chassis terminal for a ground lead or ground bus
Internal loudspeaker	Sound output when no headphones are plugged in; carries both received audio and the CW sidetone
External transmit control	Without a hand microphone, menu item 2 can turn the RIT key into a PTT key, or the radio can be keyed remotely over Bluetooth or CAT
Bluetooth	Internal Bluetooth module (power switched by menu item 1); carries CAT control and supports PTT and phone transmission over the link

13 Computer control (CAT)

The radio connects to a computer either through the USB-C serial interface or over Bluetooth. The command format is compatible with the Kenwood TS-590S (model identification reply ID021) and every command ends with a semicolon ";". The serial parameters are fixed at 115200 baud, 8 data bits, no parity, 1 stop bit. The commands in regular use are listed below.

Table 17 Commonly used CAT commands (Kenwood compatible)

Command	Function	Read / write
ID	Model identification, answers ID021	Read
IF	Complete transceiver status (frequency / mode / VFO, etc.)	Read
FA / FB	VFO-A / VFO-B frequency (11 digits, Hz)	Read / write
FR	Select VFO-A (0), VFO-B (1) or channel mode (other)	Read / write
MD	Operating mode (1=LSB, 2=USB, 3=CW-L, 4=FM, 5=AM, 7=CW-U; repeating 1 / 2 selects the data mode)	Read / write
MC / MR	Channel number (00-99) set / channel data read	Read / write
AG	Audio (AF) volume	Read / write
SM	S-meter reading (receive) / power-meter reading (transmit)	Read
RM	SWR meter reading	Read
RT	RIT on / off	Read / write
RD / RU	RIT / XIT frequency down / up (0-99999 Hz)	Write
RC	Clear RIT / XIT offset	Write
KS	CW keying speed (about 4-60)	Read / write
RX / TX	Remote receive / transmit switching (CAT PTT)	Write
BD / BU	Band select (0-10), with automatic sideband choice for the band	Write
PS	Power status query (answers PS1)	Read
FV	Firmware version (answers FV1.00)	Read
Others	Status-query commands such as AI / BY / FT / FL / DA / FS / NB / NR / PA / VX / BC / SQ / IS / SD / AN / TS / MF / NT / VG / VD / PC / MG / NL / QR / TO / TN / CT / CN / CA / KY, which answer in fixed format so that host software can poll them	Read

The Bluetooth link uses exactly the same command set as the wired serial port. In the radio-control software, choose Kenwood TS-590S as the model and 115200 baud.

14 Other parameters

Table 18 Other parameters

Item	Parameter
Display	320 x 240 colour TFT LCD, with key backlight
Panel controls	11 function keys + 2 push-encoders (AF knob, MAIN frequency knob)
Long-press threshold	A key held for about 1 second counts as a long press; storing a channel requires about 2 seconds
Key backlight	Timed (about 10 seconds after the last operation) or permanently on; menu item 9
Menu auto-exit	Menu item 0, 5-100, default 30 (the countdown is shown on screen)
Key lock	Held on and off with the F key; when locked, all keys except F, and all knobs, are ignored
Power-off	Hold the PWR key; the screen shows a farewell message, the audio is muted, and the radio shuts down after about 2 seconds
Parameter storage	All settings, frequencies, modes, channels and calibration values are saved automatically and survive power removal
Hardware adaptation	Current and earlier hardware versions are recognised at start-up; the number of front-end ATT states follows automatically

Part Two - Operating Instructions

1 Panel controls and key numbers

The front panel carries 11 function keys and 2 push-encoders - 13 controls in all. Throughout this manual they are referred to as K1-K13, and these numbers correspond exactly to the red circles in Figure 3.



Figure 3 Key numbers of the panel controls (K1-K11 are the function keys, K12 is the AF knob, K13 is the MAIN frequency knob; both knobs can also be pressed)

The following conventions are used throughout: **short press** means press and release at once; **long press** (or **hold**) means press and keep pressed for about one second, until the display changes; **hold and turn** means keep the knob pressed while rotating it. A few operations, such as storing a channel, require about two seconds.

Table 19 Overview of the panel controls

Key	Marking	Short press	Long press
K1	SSB	Enter SSB and cycle the sidebands: LSB - USB - LSB-D (data) - USB-D (data)	Phone / data / AM / FM modes: cycle the digital noise reduction NR; CW mode: switch between CW-L and CW-U
K2	M.V	Switch between VFO mode and channel (M) mode	Phone / data / AM / FM modes: pulse blanker NB on / off; CW mode: CW automatic decode on / off; in channel mode: enter channel storage
K3	CW	Enter CW and cycle the keyer type: CW M (straight key) - CW A (Iambic A) - CW B (Iambic B)	In CW mode, swap the dot / dash paddle directions
K4	AM.FM	Enter and cycle the AM / FM modes	Switch between the transceiver and receiver (broadcast) states
K5	RIT	RIT receive tuning on / off (if menu item 2 sets this key to PTT, it becomes the transmit key instead)	In CW mode, keyer practice on / off
K6	A/B	Switch VFO-A / VFO-B	Switch between same-frequency and split operation

K7	IF.ATT	Cycle the receive front end: pre-amplifier - through - attenuator	Switch between AGC (automatic gain) and MGC (manual gain)
K8	PWR	Open the transmit power (POW) window	Switch the radio on / off
K9	MENU	Enter / leave the menu	-
K10	FILTER	Cycle the receive filter bandwidth for the current mode	Receive I/Q amplitude trim window
K11	F	Enter band-select state (K1-K10 then act as band shortcuts; press again to leave)	Key lock on / off
K12	AF knob	Rotate for volume; short press cycles the multi-function adjustment windows	Open the squelch SQL window
K13	MAIN knob	Rotate to tune; short press swaps between the 1 kHz and 100 Hz steps; in the menu it means "enter / leave this item"	Hold and turn: cycle the nine tuning steps

Note: the same key can mean different things in different modes (the long-press functions of K1, K2 and K5, for example); the table lists them mode by mode. The mode box at the upper left of the screen (item 2 in Figure 4) tells you which mode you are in.

2 Switching on and off; start-up key combinations

2.1 Normal start-up and shut-down

- **Switching on**: hold PWR (K8) for about one second. The start-up screen shows the model, the date and the firmware version, after which the radio restores the state it was in when it was last switched off (band, frequency, mode, VFO / channel and all settings).
- **Switching off**: hold PWR (K8). The screen shows the farewell message "73", the audio is muted, and the radio powers down after about two seconds. Every parameter was already saved while the radio was running, so nothing else has to be done.

2.2 Start-up key combinations

The start-up key combinations enable transmit permission and enter the calibration states. **Method: press and hold the stated key first, then switch the radio on (hold K8); release the key once the message appears on screen.** The result is saved automatically and still applies at the next start-up.

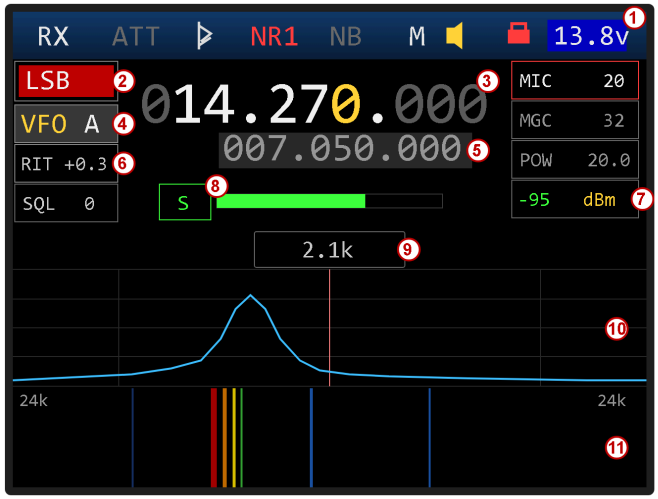
Table 20 Start-up key combinations

Hold at start-up	Screen message	Function
K2 (M.V)	60m RX/TX	Toggle the 60 m band between "may transmit" and "receive only"
K10 (FILTER)	6m RX/TX	Toggle the 6 m band between "may transmit" and "receive only"
K9 (MENU)	Development mode	Enter developer mode, which adds all calibration items to the menu (see Table 29)
K7 (IF.ATT)	Rx amp/phase adjust	Enter the receive image-rejection calibration (amplitude AMP / phase PHASE)

Receive image-rejection calibration: once inside, a short press of the frequency knob K13 switches between amplitude (AMP) and phase (PHASE); rotating K13 changes the value. Watch the image position on the spectrum and adjust the value until that spur is smallest. The calibration is stored separately for each band.

> Whether transmit is permitted on the 60 m and 6 m bands must follow the scope of your own amateur licence; a band whose permission is not enabled receives only. See the regulatory notes in section 21.

3 The main screen



1 Status bar: RX/TX, front-end ATT, Bluetooth, noise reduction (NR), pulse blanker (NB), AGC speed, speaker, key-lock, supply voltage
2 Mode box
3 Main frequency (yellow digit = the digit the current tuning step acts on; the last three Hz digits are shown dim)
4 VFO A/B indicator, or channel number in channel mode
5 Sub-VFO frequency
6 RIT shift
7 Parameter column: MIC / MGC / POW (SIVOL / SITON / KEYSP in CW); the cell below shows level in dBm on receive, output power in W on transmit
8 S-meter bar (power bar on transmit, marked S or Po at the left end)
9 Receive filter bandwidth
10 Real-time spectrum (red centre line = RX frequency; side lines +/-20 kHz)
11 Waterfall (corners show half span 24k / 12k; with CW decode on, this area shows decoded characters, measured keying speed and signal level)

Figure 4 The main screen and its areas (the illustration only shows where each area is and what it means; the actual contents and colours change with the mode and the state)

Table 21 Screen areas and their contents

No.	Area	Contents and how to use it
1	Status bar (top line)	From left to right: receive / transmit state (RX / TX), front-end attenuator ATT, Bluetooth symbol, noise-reduction NR indication, pulse blanker NB indication, AGC speed (S / M / F), loudspeaker and squelch state, key lock, supply voltage. **An indication is dim grey when the function is off and turns red when it is on**, so the colours alone tell you the present state.
2	Mode box	Upper left, white on red; shows the current mode (LSB / USB / LSB-D / USB-D / CW-L / CW-U / AM / FM).
3	Main frequency	The working frequency in large figures: 9 digits and 2 decimal points, the rightmost three being the Hz digits (shown dim). **One digit is yellow: it is the digit the current tuning step acts on** - look at the yellow position to know which digit each click of the knob changes, so the wrong order of magnitude is avoided.
4	VFO / channel indication	Shows VFO-A, VFO-B or the present channel number (the channel index in M mode).
5	Sub frequency	The frequency of the other VFO (or the other half of a channel pair), used to compare the receive and transmit frequencies. In split transmission the main and sub positions swap automatically.
6	RIT offset box	With RIT on, the present receive offset is shown (for example +0.30 / -0.50 kHz); with RIT off no value is shown.
7	Right-hand parameter column	Three small windows: in phone / data modes MIC (microphone gain), MGC (manual gain), POW (transmit power); in CW mode SIVOL (sidetone volume), SITON (sidetone pitch), KEYSP (keying speed). The cell below them shows the signal level in dBm on receive, and the output power in W together with the SWR on transmit.
8	Signal bar / power bar	On receive the bar is marked "S" at its left end and follows the signal strength; on transmit it is marked "Po" and shows the output power. The SWR cell appears at the same time as transmission starts.
9	Filter bandwidth box	The receive digital filter bandwidth in use (for example 2.4k or 500), which changes with a short press of K10.

10	Real-time spectrum	The audio spectrum centred on the receive centre frequency. The red vertical in the centre is the current frequency and the side verticals mark +/-20 kHz. The shape of the trace shows where signals are, how wide they are and whether there is interference.
11	Waterfall	The scrolling history of the spectrum; the lower-left and lower-right corners show the half span (24k or 12k). When CW decode is on, this area is used instead for the decoded text, the measured keying speed and the signal level.

Use the S bar and the spectrum area together: the S bar gives the overall strength figure, while the spectrum and waterfall show where a signal is, how wide it is and whether anything is interfering - the two complement each other.

4 Tuning and selection of the tuning step

- **Ordinary tuning**: rotate the frequency knob K13; the present VFO (or channel) frequency follows. Crossing the edge of the band switches to the neighbouring band automatically.
- **Swapping the common steps**: short press K13 to switch between 1 kHz and 100 Hz, which suits everyday QSY.
- **Selecting from all nine steps**: keep K13 pressed and turn it; the display runs through 1 Hz, 10 Hz, 100 Hz, 1 kHz, 10 kHz, 100 kHz, 1 MHz, 5 kHz and 9 kHz. Release the knob at the step you want and the radio returns to tuning.
- **Alignment**: when the step changes, the frequency is aligned to a multiple of that step automatically, so a remainder such as 14.270.037 cannot occur.
- **Watch the yellow digit**: the yellow digit in the main frequency is the step digit. On the higher bands (10 m, 6 m) it is worth switching to the 100 Hz or 10 kHz step first.
- **Receiver (broadcast) state**: medium wave defaults to a 9 kHz step and the short-wave broadcast bands to a 5 kHz step, so a station can be tuned directly on its channel spacing.
- **Fast scanning**: menu item 6 (ENCODE_FREQ) sets the multiplier applied when the knob is turned quickly (0-5, default 1); the larger the value, the more the frequency jumps per click.

5 Selection of the operating mode

- **SSB (K1)**: a short press enters single sideband and cycles LSB - USB - LSB-D - USB-D. On the first entry the sideband is chosen from the frequency: USB above 9.5 MHz, LSB below it.
- **CW (K3)**: a short press enters CW and cycles the keyer type CW M (straight key) - CW A (Iambic A) - CW B (Iambic B). In CW mode, holding K1 (or cycling it) selects the receive sideband CW-L or CW-U.
- **AM / FM (K4)**: a short press cycles AM and FM. AM suits monitoring aviation and the medium- and short-wave AM broadcast bands; FM suits operators who prefer the FM style of working, although this radio covers the 6 m band and the HF short-wave bands rather than the 145 / 430 MHz bands.
- **Data modes (LSB-D / USB-D)**: the receive bandwidth is fixed at 3.0 kHz and the transmit power limit is 10 W, for use with computer digital-communication software; the only difference from the corresponding LSB / USB mode is the bandwidth and the power limit.
- The mode is stored with the frequency and the channel, so switching VFO or recalling a channel restores the mode that belonged to it.

> The audio character changes noticeably when the mode changes (CW is much narrower, data much wider); this is normal. If nothing can be heard after a change, check the volume and the squelch first (see section 8.1).

6 Band selection; transceiver / receiver states

6.1 Entering band select

A short press of F (K11) enters the band-select state, in which K1-K10 act as band shortcuts; the figure printed beside each key is the start frequency of the band it selects (MHz). Press F again to leave. While band select is active the screen states so clearly.

Table 22 Band shortcuts in transceiver state

Key	K1	K2	K3	K4	K5	K6	K7	K8	K9	K10
Band	80 m	60 m	30 m	20 m	17 m	40 m	15 m	12 m	10 m	6 m
Marking (MHz)	3.5	5	10	14	18	7	21	24	28	50

Table 23 Band shortcuts in receiver (broadcast) state

Key	Broadcast bands cycled
K1	MW - 120 m - 90 m - 75 m - 60 m
K2	49 m - 41 m - 31 m
K3	25 m - 22 m - 19 m
K4	16 m - 13 m - 11 m - 10 m

6.2 Switching between transceiver and receiver state

Holding AM.FM (K4) switches between the transceiver state (amateur bands, transmit available) and the receiver state (medium wave and short-wave broadcast, receive only). The frequencies, VFO contents and the 100 channels of the two states are **stored separately** and do not affect one another; each state restores its own settings when it is selected.

6.3 Frequency region

Menu item 10 (ITU_REGION) decides the band edges in transceiver state: 1 = China (region 1), 2 = US (region 2), 3 = Europe (region 3, factory default). Changing the region changes the upper and lower limits of every band and the automatic sideband choice; see Tables 1 and 2 in Part One.

7 Filter selection

- A short press of FILTER (K10) cycles the receive bandwidth for the current mode; the filter box on screen (item 9 in Figure 4) shows the value as it changes.
- **CW**: 50 / 100 / 200 / 300 / 500 / 800 Hz, six selections. Use 500-800 Hz when the other station keys fast or there is inter-symbol interference, 200-300 Hz when interference rejection matters most, and 50-100 Hz for very weak signals (which then demands more accurate frequency alignment).
- **SSB**: 2.1 / 2.4 kHz in everyday use. The narrower settings reject interference better, the wider ones sound more natural. For AM broadcast listening 9 kHz can be selected; for FM working, 5 kHz.
- **Data modes**: fixed at 3.0 kHz, not adjustable.
- The transmit filter bandwidth is set separately by menu item 7 (TX_FILTER, 0-5, six selections, default 4 which is about 2.7 kHz); it has nothing to do with the receive bandwidth.
- Holding K10 opens the receive I/Q amplitude trim window (200-2000, default 1000). Leave it alone in normal use; it is only adjusted together with the image calibration (see sections 2.2 and 19).

8 The AF knob and the multi-function adjustment windows

Rotating the AF knob (K12) adjusts the volume. A short press of the knob cycles through the adjustment windows; the selected window is highlighted in the right-hand parameter column (item 7 in Figure 4), and rotating the knob then changes that window's value. About three seconds after the last operation the window closes and the knob returns to volume control.

8.1 Phone / data / AM / FM modes

Table 24 AF-knob multi-function windows (phone-type modes)

Window	Meaning	How to open it	Range and use
VOL	Audio volume	Default	Rotating adjusts it directly; headphones are recommended at night
MIC	Microphone gain	Short press K12 until MIC is selected	1-99, default 20. While transmitting, speak normally and turn the knob until the power meter swings properly without the speech distorting
MGC	Manual IF gain	Hold K7 to enter MGC first, then short press K12	1-32. Sets the receive gain by hand, which avoids AGC pumping on strong signals
POW	Transmit power	Short press K8 (PWR), or short press K12	0.1-20.0 W on the HF bands; maximum 5.0 W on 6 m; maximum 10 W in the data modes. Remembered per band
SQL	Squelch threshold	Hold K12	0-99, 0 = off. Set it just high enough that the background noise is silenced, but the audio still opens normally on a signal

Cycle order: with noise reduction off, a short press of K12 switches between VOL and MIC; once MGC is in use the order is VOL - MIC - MGC - POW.

8.2 CW mode

Table 25 AF-knob multi-function windows (CW mode)

Window	Meaning	Range / default	Tip
SIVOL	Sidetone volume	0-32, default 1	How loud the sidetone is while you are keying; reduce it when using headphones
SITON	Sidetone pitch (beat frequency)	500-990 Hz in 10 Hz steps, default 650 Hz	It also fixes where received signals appear in pitch: tune so that the other station lands on this note, which makes receiving and transmitting sound the same
KEYSP	Keying speed	5-50 WPM, default 20 WPM	The dot / dash speed of the automatic keyer; start at 5-10 while learning

Holding K12 opens the squelch SQL window in every mode. In CW mode the POW window can also be reached with a short press of K8 to adjust the power.

9 Pre-amplifier / attenuator and AGC / MGC

- **Front-end state (short press K7)**: cycles "pre-amplifier (AMP) - through - attenuator (ATT)". Use the attenuator or the through state on strong signals, near-by stations or heavy atmospheric noise; use the pre-amplifier for very weak signals. The ATT mark in the status bar shows that attenuation is applied. Earlier hardware switches between the two states "pre-amplifier / attenuator", and the radio adapts to it at start-up by itself.

- **AGC / MGC (hold K7)**: switches between automatic and manual gain. With AGC the radio follows the signal strength on its own, which is the recommended everyday state. With MGC you set the IF gain by hand in the right-hand parameter column (1-32), which suits very weak CW listening, or any case where the gain should stay fixed while signals fade (signal comparison, recording).
- **AGC speed (menu item 5)**: 0 slow / 1 medium (default) / 2 fast. Slow suits weak signals and CW, because the background is quieter and signals stand out more; fast suits work that needs the gain to follow signal changes quickly.
- **AGC start level and S-meter calibration**: developer menu items, set at the factory and not needed in everyday use.

> If the received audio seems to "breathe" (pumping), try the attenuator together with slow AGC first; only if that is not good enough should MGC be set by hand.

10 Noise reduction (NR) and pulse blanker (NB)

These functions improve what the receiver sounds like: **noise reduction (NR) works on continuous, steady background noise** (mains and power-supply hum, fans and engines, induced line noise), while **the pulse blanker (NB) works on sudden pulses** (the clicking from ignition systems, sparking, appliance switches and relay action). Both can be on at the same time, and they work best together.

10.1 The NR levels and how to use them

Operation: in the phone, data, AM and FM modes, hold SSB (K1). The levels cycle **off - NR1 - NR2 - NOT - off**, and the NR indication in the status bar (item 1 in Figure 4) changes with them: **when reduction is off the indication reads "NR" in dim grey; when it is on, NR1 / NR2 / NOT appear in red**. In CW mode, holding K1 changes the sideband instead of the noise reduction.

Table 26 NR levels and where each one is used

Indication	Role	Where it is used	What it sounds like, and what to watch for
(off, dim NR)	No reduction	The signal is clear and the background clean, or you need to judge whether a signal is genuinely weak	The original audio, the most natural sound
NR1	Mild reduction	Ordinary background noise, where you want quietness without losing speech intelligibility	The recommended first choice for everyday work; speech stays natural
NR2	Deep reduction	Stronger, fairly steady noise (in a vehicle, near an inverter, hum induced on a long feedline)	Quieter still, but speech takes on a slightly "electronic" character; step back one level if the signal is weak
NOT	Automatic notch	A steady single-tone interference in the background: a fixed whistle, a carrier, a howl, a constant hum	Only that one tone is removed and the rest of the audio is left alone; suits the case where everything else is clean except one disagreeable tone

The noise reduction of this radio works adaptively: once a level is selected the radio follows the noise by itself, and there is no "strength", "delay" or similar extra value to set. Choosing the level according to the kind of noise is all that is needed; there is no reason to keep adjusting parameters.

Recommendations

1. Listen first to find out what kind of noise it is: **continuous hiss or hum** - use NR1, and move to NR2 if it is still noisy; **one fixed tone** - use NOT; **both** - use NOT to remove the tone first, then choose NR1 or NR2 as needed (the NOT level of this radio applies noise reduction as well).
2. On weak signals, or when you need to decide whether a signal is really weak, switch the reduction off briefly and compare - any reduction affects the natural sound of a weak signal to some degree.
3. A change of level takes effect at once; nothing has to be confirmed or saved, and the level you chose is still there after the radio has been switched off and on again.
4. The reduction is applied to the received audio of CW, SSB and the data modes; AM and FM are not reduced (their bandwidth and noise behaviour differ). Although the level cannot be changed while in CW mode, the level previously chosen still acts on the CW audio.

10.2 How to use the pulse blanker (NB)

Operation: in the phone / data / AM / FM modes, hold M.V (K2) to turn it on or off. The NB indication changes from dim grey to red when the blanker is on.

Table 27 Combining NB and NR

Type of interference	Suggested setting	Comment
Clicking from ignition or appliances	NB on	Enable NB first; only after the pulses are clearly reduced should NR be considered
Continuous background noise (hiss, hum)	NR1 or NR2	Nothing to do with NB; leave it to NR
Pulses together with continuous noise	NB on + NR1 / NR2	The two do not conflict, and together they give the best result
Steady single-tone interference	NR = NOT	Add NB as well if pulses are present too
Atmospheric noise in a thunderstorm	NB on, front end set to through or attenuator	With very strong atmospheric pulses the pre-amplifier only makes the overload worse

In CW mode, holding K2 is the CW decoder switch, not the NB switch - to turn NB on or off while receiving CW, change to a phone or data mode first.

11 RIT shift and the second function of the RIT key

- **Turning RIT on**: short press RIT (K5); the status bar shows RIT and the RIT box (item 6 in Figure 4) shows the present offset. While it is on, rotating the frequency knob **changes the receive frequency only and leaves the transmit frequency untouched**, so the other station can be brought to a comfortable pitch without moving your own transmission.
- **Range and increment**: +/-1200 Hz in 10 Hz steps. A further short press of K5 switches RIT off and returns the offset to zero.
- **Used as the transmit key**: after menu item 2 (RIT_PTT) is set to 1, K5 becomes the PTT key (press to transmit, release to return to receive), which allows transmitting with neither a hand microphone nor a foot switch.
- **XIT from the computer**: transmit incremental tuning (XIT) is available through the CAT commands; there is no panel key for it.

12 VFO, channels, split operation and channel storage

12.1 VFO and channel modes

- A short press of M.V (K2) switches between VFO mode and channel (M) mode; the indication area (item 4 in Figure 4) shows VFO-A / VFO-B or the channel number.
- A short press of A/B (K6) switches between VFO-A and VFO-B. The two VFOs can hold different frequencies and different modes, which makes comparing them easy.
- In VFO mode, rotating K13 tunes the frequency; in channel mode it selects the channel number (0-99).

12.2 Same-frequency and split operation

Holding A/B (K6) switches between same-frequency and split operation. In the split state one VFO serves as the receive frequency and the other as the transmit frequency (receive on A, transmit on B); during transmission the main and sub frequency positions swap automatically and the status bar indicates the state. This is what is needed for a repeater with separate input and output frequencies, or for any "listen on A, transmit on B" working. Hold the key again to return to same-frequency operation.

12.3 Storing and recalling a channel

1. Short press M.V (K2) to enter channel (M) mode;
2. Hold M.V (K2) to enter channel editing;
3. Rotate the frequency knob K13 to select the channel number to be written (0-99);
4. Keep M.V (K2) held for about two seconds and release it when the storage message appears - the frequency, mode, filter, tuning step, RIT, same/split state, transmit power, squelch, CW parameters, NR / NB, front-end state, MIC gain, transmit filter and the rest of the state are then written into that channel together;
5. **Recall**: in channel mode, rotate K13 to the channel number and the radio immediately restores every item stored in it.

Channels are stored separately for the transceiver and receiver states, so one never overwrites the other. To change what a channel holds, recall it, set the radio as you want it, and write it again following the steps above.

13 CW transmitting and receiving

13.1 Connecting the key and choosing the keyer type

- **Straight key**: connect it to the CW.KEY socket, short press CW (K3) until CW M is selected; press the key to transmit, release to return to receive.
- **Paddle**: connect the dot and dash sides to the two keyer lines, then short press K3 until CW A (Iambic A) or CW B (Iambic B) is selected. When both paddles are pressed together, dots and dashes are generated alternately according to the Iambic rule chosen.
- **Dots and dashes reversed**: in CW mode, hold K3 to exchange the two paddle lines; the setting is saved automatically.

13.2 Sidetone, keying speed and beat frequency

- In CW mode, short presses of K12 bring up SIVOL (sidetone volume 0-32), SITON (sidetone pitch 500-990 Hz in 10 Hz steps) and KEYSP (keying speed 5-50 WPM) in turn; rotate the knob to adjust them. See Table 25.
- The sidetone pitch is also the receive beat position: changing the pitch moves the received signals with it. Set SITON first to a pitch that is comfortable to you (650 Hz by default), then use RIT or tuning to place the other station on it.

- The keying speed defaults to 20 WPM; beginners may start at 5-10 WPM and increase it with practice.

13.3 Transmitting and keyer practice

- **Hold-over delay**: menu item 4 (CW_DELAY), 0-99, each count being 10 ms, default 5 (50 ms). This is how long the transmitter stays keyed after the key is released; a suitable setting avoids breaking between letters and words and the click that comes with it. Set to 0 for the fastest break-in.
- **Practice mode**: in CW mode, hold RIT (K5) to open practice mode. Keying then produces sidetone only, with no transmission, so sending can be practised quietly; hold the key again to close it. The screen indicates the practice state.
- During transmission the screen shows the output power and the SWR; the start and end of each element are ramped in power to remove key clicks.

13.4 CW decode

- **Turning it on**: set menu item 14 (CW_DEC) to 1, or hold M.V (K2) while in CW mode.
- Once on, the waterfall area at the bottom of the screen shows the decoded text instead, together with the measured keying speed (WPM) and the signal level - useful for checking the other station's rate and your own copy.
- Decoding adapts automatically to keying speeds of roughly 10 to over 50 WPM and tolerates about +/-150 Hz of beat error. Occasional wrong characters on weak, fading signals or from uneven straight-key sending are normal; tuning the signal in properly and narrowing the receive filter improve it noticeably.

14 Phone and data transmission

14.1 Checks before transmitting

1. Connect an antenna that matches the band and has a normal SWR; make sure the 12 V d.c. supply has enough capacity (centre positive; see Table 16 in Part One).
2. Select the correct mode and a **frequency on which transmission is permitted** (the transmit sub-bands are in Table 2 of Part One).
3. Short press PWR (K8) to open the POW window and rotate K12 to set the power for this band: 0.1-20.0 W on the HF bands, 5.0 W maximum on 6 m, 10 W maximum in the data modes. The power is remembered band by band.
4. For the 60 m and 6 m bands the transmit permission must first be enabled with the start-up key combinations (Table 20).
5. Try transmitting at low power first and confirm that the SWR is normal before increasing it.

14.2 SSB / AM / FM transmission

- Press the PTT on the hand microphone to transmit; releasing it returns the radio to receive. Transmission can also be started by PTT over the Bluetooth link, by a CAT command, or by assigning K5 as the PTT key (menu item 2) and pressing it. Note that the socket marked PTT.OUT is a **transmit-control output** (for switching an external amplifier or antenna); it cannot be used as a PTT input.
- **Microphone gain**: short press K12 to open the MIC window, then, while transmitting, speak normally into the microphone and turn the knob until the output power swings evenly with the speech without overload or distortion. The default is 20, and 15-30 usually suits.
- **Speech compression**: menu item 8 (TX_AF_COMP, 1-99, default 10). More compression raises the average output and makes the signal stand out better under poor propagation, but too much makes the speech dull and

invites howl-around; keep the default or increase it in small steps.

- **Transmit bandwidth**: menu item 7 (TX_FILTER, 0-5, default 4 which is about 2.7 kHz).
- During transmission the screen shows the output power (W), the SWR and the supply voltage continuously. Stop transmitting at once if the SWR is clearly high, and check the antenna and feedline; a large drop in supply voltage means the power supply is not up to the load.
- AM and FM can transmit as well; FM reception has 5 / 10 kHz bandwidth selections.

14.3 Data-mode transmission

1. Short press K1 until LSB-D or USB-D is selected (bandwidth fixed at 3.0 kHz, power limit 10 W).
2. Connect the sound-card audio output of the computer to the MIC socket and its audio input to the headphone / loudspeaker output socket; the PTT is handled by CAT from the digital software or by an external PTT.
3. Establish the link at a lower power first, then set the MIC gain using the level meters of the digital software itself, so that overload does not distort the signal and widen the spectrum.
4. Do not leave deep speech compression enabled while transmitting in a data mode.

The USB-C interface carries the CAT control and the data-mode audio to and from the computer; the radio is powered from its d.c. socket, not from USB.

15 Spectrum and waterfall display adjustments

- **Span**: menu item 11 (SPE_W); 1 = about 48 kHz (waterfall corners marked 24k), 2 = about 24 kHz (corners marked 12k). The narrow span shows signal detail better, the wide span gives an overview of activity across the band.
- **Colour**: menu item 12 (SPE_COLOR), 0-6, seven schemes: 0 / 1 are colour gradients (filled / outline), 2-6 single-colour outline (blue / yellow / green / red / grey). Choose by personal taste and by how bright the surroundings are.
- **Display amplitude**: menu item 13 (SPE_AMP), 1-10, default 4. Increase it if signals look too low, reduce it if the background is too bright.
- **Transmit spectrum**: developer menu item 4 enables spectrum monitoring while transmitting, useful for looking at the occupied bandwidth and spurious emissions; it need not be on in everyday use.
- **Reading the display**: the red centre line is the receive frequency, so whether a signal appears to its left or right corresponds to the lower or upper sideband selected. The density of the tracks in the waterfall history shows whether a frequency carries regular activity, which helps decide where to move to.

16 The menu system

16.1 How to operate the menu

1. Short press MENU (K9) to enter the menu;
2. Rotate the frequency knob K13 to select a menu item (previous / next);
3. Short press K13 to enter that item's value, then rotate K13 to change it;
4. A further short press of K13 returns to item selection; a short press of K9 saves and leaves the menu;
5. The menu closes by itself: after the last operation it exits automatically at the time set in menu item 0, with a countdown on screen. Every change is saved at the moment it is made.

16.2 User menu (15 items, numbered 0-14)

Table 28 User menu

No.	Name	Meaning	Range	Default
0	MENU	Menu auto-exit time	5-100	30
1	BLUETOOTH	Bluetooth module power (0 off / 1 on)	0-1	1 (on)
2	RIT_PTT	Second function of the RIT key (K5): 0 = RIT, 1 = PTT transmit key	0-1	0
3	IF 1	Digital IF frequency (Hz); normally left at the default	20000-28000	24000
4	CW_DELAY	CW hold-over delay (10 ms per count)	0-99	5 (50 ms)
5	AGC_SPEED	AGC recovery speed: 0 slow / 1 medium / 2 fast	0-2	1
6	ENCODE_FREQ	Multiplier applied when the frequency knob is turned quickly	0-5	1
7	TX_FILTER	SSB transmit filter selection (6)	0-5	4
8	TX_AF_COMP	Speech compression strength	1-99	10
9	KEY_BL	Key backlight: 0 timed / 1 always on	0-1	0
10	ITU_REGION	Frequency region: 1 China / 2 US / 3 Europe	1-3	3
11	SPE_W	Spectrum span: 1 = about 48 kHz / 2 = about 24 kHz	1-2	1
12	SPE_COLOR	Spectrum / waterfall colour scheme	0-6	0
13	SPE_AMP	Spectrum display amplitude	1-10	4
14	CW_DEC	CW decode on / off	0-1	1 (on)

16.3 Developer / calibration menu (28 items)

After developer mode has been entered by holding MENU (K9) at start-up, the following items are added to the user menu. They concern frequency accuracy, output power, sideband rejection and the image figure; the radio is calibrated at the factory, so **they should not be changed except by qualified service personnel**. Any change is saved automatically.

Table 29 Developer / calibration menu

No.	Name	Meaning	Range	Factory value
0	POW_CORRE	Transmit power calibration coefficient	50-100	70
1	AGC_STARE	AGC start level	2-127	64
2	S_CORRECT	S-meter calibration	0-63	5
3	TCXO	Frequency reference (Hz)	24.900000-27.100000 MHz	25.000000 MHz
4	SPE_DISPALY	Transmit spectrum display on / off	0-1	0
5	RX_AMP_MA	Receive I/Q amplitude balance: 0 manual / 1 automatic	0-1	0
6	ALC_START	Microphone ALC start threshold	0-15	15
7	ALC_MIC	ALC maximum gain	0-63	32

8 / 9	TX_AMP / TX_PHASE (80 m)	Per-band transmit I/Q amplitude / phase calibration	Amplitude 500-2000 Phase 50-200	1000 / 99
10 / 11	TX_AMP / TX_PHASE (60 m)	As above, 60 m		
12 / 13	TX_AMP / TX_PHASE (40 m)	As above, 40 m		
14 / 15	TX_AMP / TX_PHASE (30 m)	As above, 30 m		
16 / 17	TX_AMP / TX_PHASE (20 m)	As above, 20 m		
18 / 19	TX_AMP / TX_PHASE (17 m)	As above, 17 m		
20 / 21	TX_AMP / TX_PHASE (15 m)	As above, 15 m		
22 / 23	TX_AMP / TX_PHASE (12 m)	As above, 12 m		
24 / 25	TX_AMP / TX_PHASE (10 m)	As above, 10 m		
26 / 27	TX_AMP / TX_PHASE (6 m)	As above, 6 m		

17 Key lock, backlight, headphones and Bluetooth

- **Key lock**: hold F (K11) to switch it on and off. When locked, the LOCK indication in the status bar flashes and, apart from the F key and switching the radio off, both the keys and the tuning knob are ignored - which prevents accidental changes while the radio is being carried or cleaned. Hold F again to release it.
- **Key backlight**: with menu item 9 set to 0, the backlight comes on with each operation and goes out about ten seconds afterwards (saving power); set to 1 it stays on. For field or night use, 0 is recommended.
- **Headphones and external loudspeaker**: inserting a plug in PHONES mutes the internal loudspeaker, and removing it restores the loudspeaker. An external loudspeaker can be connected to EXT.SP. Headphones are recommended at night or in noisy surroundings. Note that the headphone socket also carries the audio path in the data modes, so stop transmitting before plugging in or unplugging.
- **Bluetooth**: menu item 1 switches the power of the Bluetooth module (**enabled at the factory**). Once the link is connected, the Bluetooth symbol appears in the status bar. The link carries exactly the same computer-control command set as the wired serial port, and also supports Bluetooth PTT and phone working. Pairing is done as with any ordinary Bluetooth serial device, by searching for and connecting to the radio from the computer or telephone. Switch the item off to save power if Bluetooth is not used for a long time.

18 Computer connection and CAT operation

1. Connect the USB-C socket of the radio to the computer with a data cable, or use the Bluetooth link; the serial parameters are 115200 baud, 8 data bits, no parity, 1 stop bit.
2. In the radio-control, logging or digital software, choose **Kenwood TS-590S** as the radio model (this radio identifies itself as ID021); commands end with a semicolon ";".
3. Available from the computer: frequency read and write (FA / FB), mode change (MD), VFO and channel selection (FR / MC / MR), volume (AG), S-meter and power readings (SM), SWR (RM), RIT / XIT (RT / RD / RU / RC),

CW keying speed (KS), remote receive / transmit (RX / TX, i.e. CAT PTT), band select (BD / BU), firmware version (FV, answered with FV1.00) and more; the complete list is in Table 17 of Part One.

4. Digital-mode software uses CAT for PTT and the sound-card paths for the audio (see section 14.3).
5. Do not connect the wired and the Bluetooth link to the same control program at the same time, or the two sets of commands will conflict; disconnect one before switching to the other.

19 Calibration functions

- **Receive image rejection**: entered by holding IF.ATT (K7) at start-up. Use K13 - short press - to switch between AMP (amplitude) and PHASE (phase), rotate to adjust, and bring the spur at the image position on the spectrum to its minimum; stored separately per band.
- **Frequency reference**: developer menu item 3 (TCXO), a 25 MHz temperature-compensated crystal oscillator adjustable over 24.900000-27.100000 MHz. Trim it against a standard frequency source (a standards transmitter, a frequency counter or a calibrated receiver) until the indicated frequency agrees with the actual one.
- **S meter / AGC start level**: developer menu items 2 and 1.
- **Transmit I/Q amplitude and phase**: developer menu items 8-27, calibrated band by band for sideband rejection and carrier leakage; this needs a spectrum analyser or a receive comparison and belongs to production or service.
- **Transmit power**: developer menu item 0, corrected with a power meter. In everyday use the output power of the present band can also be trimmed from the POW window while transmitting, using K12.

20 Factory default parameters

The table below lists the main defaults that apply the first time the radio is powered up, when the memory is still blank. Every change made afterwards is saved automatically and is restored when the radio is switched on again; to return to a default, simply set the item back by hand.

Table 30 Factory default parameters

Item	Default	Item	Default
State at start-up	VFO-A, 40 m, 7.050 MHz, LSB	ITU frequency region	Region 3 (Europe)
Digital IF	24 kHz	Frequency reference	25.000000 MHz
Tuning step	1 kHz	AGC speed	Medium (M)
SSB receive bandwidth	2.1 kHz	CW receive bandwidth	500 Hz
AM bandwidth	6 kHz	FM bandwidth	5 kHz
CW keyer type	Straight key (CW M)	CW keying speed	20 WPM
CW sidetone pitch	650 Hz	CW sidetone volume	1
CW hold-over delay	50 ms	CW decode	On
MIC gain	20	Speech compression	10
Transmit SSB bandwidth	Selection 4 (about 2.7 kHz)	HF transmit power	20.0 W
6 m power limit	5.0 W	Data-mode power limit	10.0 W
Squelch SQL	0 (off)	Noise reduction NR	Off
Pulse blanker NB	Off	Front-end state	Pre-amplifier in
Bluetooth	On	RIT key function	RIT (not PTT)

Key backlight	Timed	Menu auto-exit	30
Spectrum span	About 48 kHz	Spectrum colour / amplitude	0 / 4

21 Operating precautions and regulatory notes

- This is an amateur radio transceiver. Transmitting must comply with the radio regulations of the country or region concerned; the operator must hold the appropriate amateur licence and must transmit only on the amateur frequencies and at the powers covered by that licence.
- Transmit permission for the 60 m and 6 m bands is controlled by the start-up key combinations (Table 20); before enabling them, confirm that local regulations permit transmission on those frequencies. In receiver state, and on any band whose permission has not been enabled, the radio **may only receive - transmission is not permitted**.
- Always connect a serviceable antenna with a normal SWR before transmitting; **never transmit into an open circuit (with no antenna)**. Stop at once if the SWR is high, the supply voltage drops noticeably or the case becomes hot.
- Use a proper 12 V d.c. supply (centre positive) with enough current headroom for transmitting, and never reverse the polarity. During thunderstorms avoid the external antenna and disconnect the feedline.
- The limits of 20 W on CW / SSB, 10 W in the data modes and 5 W on 6 m are the normal design values, not a fault.
- Every item of the developer calibration menu is set at the factory; do not change them, or frequency accuracy, sideband rejection and the output power figure will suffer.
- When interconnecting with a computer, audio equipment and the like, take care over common bonding and level matching to avoid ground-loop noise; before establishing a data link, reduce the power first.
- Noise reduction and filtering improve the sound, but they also make a very weak signal seem less natural; when judging a weak signal, switch the reduction off briefly and compare.
- Provide ventilation during long transmissions at high power. The display is a colour TFT panel; do not leave the radio in direct sunlight or at high temperature.

Appendix - Quick-reference table

Table 31 Quick reference

What you want to do	Key / method
Switch on / off	Hold PWR (K8)
Tune the frequency	Rotate the frequency knob K13
Swap the common tuning steps	Short press K13 (1 kHz / 100 Hz)
Select from all nine steps	Hold K13 and turn it
Change SSB / data mode	Short press SSB (K1) to cycle LSB / USB / LSB-D / USB-D
Change CW mode and keyer type	Short press CW (K3) to cycle CW M / CW A / CW B
Change AM / FM	Short press AM.FM (K4)
Transceiver / receiver state	Hold AM.FM (K4)
Select a band	Short press F (K11), then a band shortcut (Tables 22, 23)
Change the receive filter	Short press FILTER (K10)
Volume	Rotate the AF knob K12
MIC / MGC / POW / sidetone / keying speed	Short press K12 to the window, then rotate (Tables 24, 25)
Squelch	Hold K12

Transmit power	Short press PWR (K8), then rotate K12
Front end: amplifier / through / attenuator	Short press IF.ATT (K7)
AGC / MGC	Hold IF.ATT (K7)
Noise reduction on / off, NR1 / NR2 / NOT	Hold SSB (K1) in the phone, data, AM or FM mode to cycle
Pulse blanker NB on / off	Hold M.V (K2) in the phone, data, AM or FM mode
What holding K1 / K2 does in CW mode	K1 = CW-L / CW-U; K2 = CW decode on / off
RIT on / off	Short press RIT (K5)
VFO / channel mode	Short press M.V (K2)
VFO-A / VFO-B	Short press A/B (K6)
Same-frequency / split	Hold A/B (K6)
Store a channel	In channel mode hold K2, select the number, then keep K2 held about 2 s
Swap CW dot / dash paddles	Hold K3 in CW mode
CW keyer practice	Hold K5 in CW mode
Enter / leave the menu	Short press MENU (K9)
Choose a menu item / change its value	Rotate K13 to choose, short press K13 then rotate to change
Key lock on / off	Hold F (K11)
Enable 60 m / 6 m transmit	Hold K2 / K10 while powering up
Developer calibration menu	Hold K9 while powering up
Receive image-rejection calibration	Hold K7 while powering up

- End -